VRP and QAOA: Qubit Allocation Strategies

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What are VRP and QAOA?

- **VRP (Vehicle Routing Problem):** given a depot (start/end) and customers, find a short tour that visits each customer exactly once. Generalizes TSP. Brute-force enumeration scales as $N!$ for N locations.
- **QAOA (Quantum Approximate Optimization Algorithm):** a hybrid quantum--classical method that encodes an objective into a quantum "cost" Hamiltonian, alternates it with a mixing Hamiltonian, and classically tunes parameters (γ, β) to favor low-cost bitstrings.

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> *We can pose VRP as a QUBO/Ising model and run QAOA to search for good (not necessarily optimal) routes.*

Allocating Qubits: Theoretical Minimum

Logarithmic Permutation Encoding

- **Idea:** Encode the order of visits as one of the N! permutations. If a register of k qubits can represent 2^k distinct states, the information-theoretic lower bound to index all tours is:

$$k_{\min} = \lceil \log_2(N!) \rceil$$

- This is minimal because any injective encoding of N! objects needs at least $\log_2(N!)$ bits/qubits. (Stirling's approximation shows $\log_2(N!) \sim N \log_2 N - 1.44N$)
- **Practical snag:** Turning a permutation index into a *local cost Hamiltonian* is hard. You end up doing a table lookup of tour lengths (or heavy classical post-processing), which undermines the benefits of in-circuit optimization.

Practical Qubit Allocation: What People Actually Use

- **One-Hot / Edge (Naive Adjacency) Encoding**

  Use one binary variable per directed edge or per city × time slot, requiring N^2 qubits for N locations.

  - Cost: Sum of chosen edge weights + penalties for constraints (degree/flow).
  - Straightforward to encode in QAOA/Ising Hamiltonian.

- **Binary Node-Visit-Time (Ranking/Position) Encoding**

  Assign each customer a small binary register storing their tour position (1…N), using $N\lceil\log_2 N\rceil$ qubits.

  - More qubit-efficient than one-hot encoding.
  - Penalize duplicate positions and out-of-range values.
  - Cost/constraints are trickier but still practical for QUBO/Ising implementation.

Encoding Method Comparison

Encoding Method	Qubit Count	Num of Qubit	Cost Function Simplicity
Naive Matrix (NxN)	N^2	High	Simple
Log Permutation	$\lceil \log_2(n!) \rceil$	Low	Complex (impractical)
Node-Visit-Time	$n \cdot \lceil \log_2(N) \rceil$	Moderate	Moderate

Trade-off: More qubits generally allow for simpler cost Hamiltonians and penalties in QAOA implementations.

Why Not Just Use the Theoretical Minimum?

- Minimal-qubit encoding packs the information tightly, but the cost function becomes a global lookup rather than a local polynomial in qubits.
- Implementing as a Hamiltonian either (i) requires precomputing all tour costs and effectively brute-forcing via post-processing, or (ii) produces highly non-local operators that are impractical on near-term hardware.
- The minimal encoding is fundamentally limited by the same combinatorial explosion as brute-force TSP.

> **In short:** One-hot and binary position encodings spend extra qubits to allow local, simple cost/penalty terms that QAOA can practically optimize.